



THROTTLE LK

SOFTWARE REQUIREMENTS SPECIFICATION

Sri Lanka's Motorbike Marketplace

Motorcycles • Scooters • Spare Parts • Modified Parts • Dealers

Document	Details
Version	1.1
Prepared by	PT Dilshan
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Product model	Web-first, API-first marketplace with planned Flutter mobile application

Commercial Business Product Documentation

Document Control

Item	Details
Product name	ThrottleLK — Sri Lanka’s Motorbike Marketplace
Document	Software Requirements Specification
Version	1.1
Prepared by	PT Dilshan
Product owner	PT Dilshan
Date	September 2026
Primary sources	ThrottleLK Business & Project Proposal; existing ThrottleLK SRS v1.0; product/technical specification and implemented monorepo where available
Reference structure	Professional SRS structure informed by the supplied AutoValueLK SRS sample; academic-specific wording removed for this commercial product

Revision History

Version	Date	Description
1.0	September 2026	Initial ThrottleLK SRS.
1.1	September 2026	Business-focused revision. Removes university/supervisor framing, standardizes Flutter as the planned mobile client, strengthens commercial, operational, security, data, SEO and launch requirements.

Requirement Status Legend

Status	Meaning
Implemented	Present in the current product/codebase and available in a real user journey.
Partial	A portion of the required data model, API, UI or workflow exists, but the full requirement is not complete.
Planned	Approved product requirement not yet delivered.
Future	Intentionally outside the first commercial release; retained for product roadmap visibility.
TBD	A business or technical decision that must be finalized before the relevant release baseline.

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1. Introduction [\[Back to TOC\]](#)

1.1 Purpose [\[Back to TOC\]](#)

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for ThrottleLK, a commercial motorcycle-first digital marketplace for Sri Lanka. It is the product baseline for business planning, development, quality assurance, operations, security review and future platform expansion.

ThrottleLK connects buyers with private sellers, approved motorcycle dealers and approved parts dealers. Users can discover motorcycles, scooters, spare parts and modified parts; sellers can publish structured inventory; and administrators can control marketplace quality, taxonomy, reports and promotions. Vehicle sales are completed outside the platform through phone, WhatsApp or in-app communication. The first commercial release does not process vehicle purchase payments, escrow, ownership transfer or physical inspection.

1.2 Document Conventions [\[Back to TOC\]](#)

- Functional requirements use a feature prefix and three-digit number, such as AUT-001, LST-001 and DLR-001.
- Nonfunctional requirements use category prefixes such as PE (performance), SE (security), UI (user interface) and OE (operating environment).
- The word “shall” denotes a mandatory requirement for the release scope in which it appears.
- Implemented, Partial, Planned and Future describe delivery status; they do not replace the requirement itself.
- TBD identifies a product, commercial or technical decision that has not yet been finalized.

1.3 Intended Audience and Reading Suggestions [\[Back to TOC\]](#)

Audience	Recommended sections
Founder / Product Owner	1, 2, 6, 8, 12, 13
Business / Commercial team	1, 2, 3, 6, 8, 12
Software developers	2, 3, 4, 7, 10, 11
QA / Test engineers	3, 5, 9, 10, 12
Operations / DevOps	2, 4, 5, 8, 11, 13
Future maintainers / partners	Whole document, especially 7, 11 and Appendix B

1.4 Product Scope [\[Back to TOC\]](#)

ThrottleLK addresses the fragmented experience of motorcycle buying, selling and parts discovery in Sri Lanka by providing one motorcycle-focused marketplace instead of relying only on general classifieds, social-media groups and individual dealer pages.

- Motorcycle- and scooter-specific structured listings, search and filtering.
- Moderated publishing for private sellers and approved dealers.
- Approved bike-dealer and parts-dealer public showrooms.
- Spare-parts and modified-parts marketplace with bike brand/model fitment.
- Favourites, comparison, saved searches, messaging, notifications and seller/dealer lead generation.
- English and Sinhala public experience.
- SEO-oriented permanent pages for listings, brands, models, categories, locations and dealers.
- Versioned REST API used by the web application and the planned Flutter mobile application.

ThrottleLK is a discovery, advertising and lead-connection marketplace. It does not provide an official vehicle valuation and does not complete ownership transfer or the motorcycle purchase transaction in the first release.

1.5 References [\[Back to TOC\]](#)

1. ThrottleLK Business & Project Proposal, PT Dilshan, September 2026.
2. Existing ThrottleLK Software Requirements Specification v1.0.
3. ThrottleLK product/technical specification and implemented monorepo where available.
4. Supplied AutoValueLK SRS v1.0, used only as a reference for professional SRS organization and presentation.

2. Overall Description [\[Back to TOC\]](#)

2.1 Product Perspective [\[Back to TOC\]](#)

ThrottleLK is a standalone commercial marketplace, not an extension of a university project or an academic prototype. The platform is web-first and API-first so the same business logic can later support a dedicated Flutter mobile application.

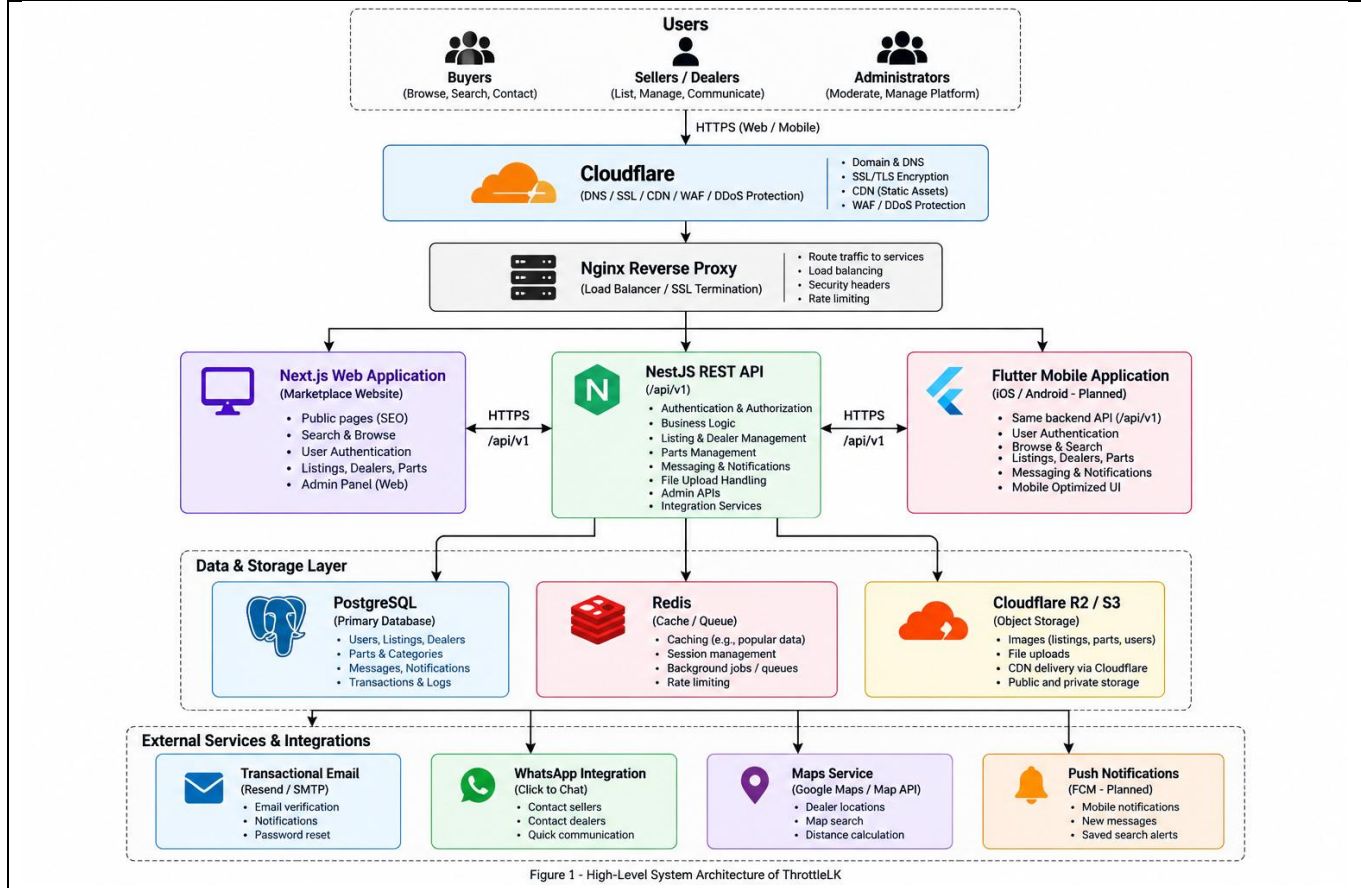


Figure 1 - High-Level System Architecture of ThrottleLK

2.2 Product Features [\[Back to TOC\]](#)

Feature	Summary	Delivery
Authentication & roles	Registration, login, JWT access/refresh, email verification, password reset and RBAC.	Implemented
Motorcycle listings	Create, edit, images, submit, moderate, pause/resume and mark sold.	Implemented
Search & browse	Motorcycle-specific filters, sorting, keyword search and SEO landing pages.	Partial
Listing detail & contact	Gallery, specs, seller information, phone, WhatsApp, inquiry, messaging and engagement tracking.	Implemented
Favourites & compare	Save bikes, price-drop notifications and compare up to three bikes.	Implemented
Saved searches	Persist named searches; matching alerts require a background worker.	Partial
Messaging & notifications	In-app conversations, notification feed and transactional email.	Partial
Bike dealers	Application, approval, showroom, inventory, documents, map and performance.	Implemented
Parts marketplace	Spare/modified parts, categories, fitment, inquiries and favourites.	Implemented
Parts dealers	Application, approval, showroom and map directory.	Implemented

Feature	Summary	Delivery
Trust & moderation	Reports, approval queues, rejection/take-down and protected data.	Implemented / Partial
Promotions	Packages, bank-slip review and homepage placements.	Implemented - manual settlement
Admin console	Users, taxonomy, listings, dealers, parts, reports and promotions.	Implemented
SEO & public content	SSR, metadata, canonicals, Open Graph, JSON-LD, sitemap and robots.	Partial
Bilingual UX	English/Sinhala routing and content.	Partial
Flutter mobile app	Buyer/seller/dealer mobile experience using the same API.	Planned
Online payments / subscriptions	Paid featured products, dealer plans and related commercial checkout.	Future

2.3 User Classes and Characteristics [\[Back to TOC\]](#)

User class	Characteristics	Primary use
Guest Buyer	No account required for basic discovery.	Browse, filter, compare locally, view listings and begin contact.
Registered Buyer	Authenticated marketplace user.	Favourites, saved searches, messages, reports, notifications and profile.
Private Seller	Individual selling one or more motorcycles.	Create and manage listings, receive leads, view performance.
Approved Bike Dealer	Business user approved by administration.	Public showroom, inventory, documents, leads and analytics.
Approved Parts Dealer	Parts business approved by administration.	Parts inventory, fitment, showroom and map presence.
Administrator	Marketplace operations and quality-control user.	Moderation, taxonomy, users, dealers, reports, promotions and platform control.
Advertiser / Partner	Future commercial stakeholder.	Homepage/category placements and future commercial campaigns.

2.4 Operating Environment [\[Back to TOC\]](#)

ID	Requirement	Status
OE-001	The product shall operate as a responsive web application in modern mobile and desktop browsers.	Implemented
OE-002	The web frontend shall use Next.js, React and TypeScript.	Implemented
OE-003	The backend API shall use NestJS and TypeScript.	Implemented
OE-004	PostgreSQL shall be the operational system of record.	Implemented
OE-005	Shared TypeScript contracts/validation shall be reused across the platform where available.	Implemented
OE-006	Production traffic shall be HTTPS and may be served behind Cloudflare and Nginx; final hosting topology is TBD.	Partial
OE-007	Redis shall support cache and coordination when configured.	Implemented
OE-008	Media shall be stored in S3-compatible object storage such as Cloudflare R2.	Implemented
OE-009	The server runtime shall use a supported Node.js production version; current target is Node.js 22+.	Implemented
OE-010	The planned mobile application shall be implemented in Flutter/Dart and consume the same /api/v1 API.	Planned

2.5 Design and Implementation Constraints [\[Back to TOC\]](#)

- The web marketplace is the first commercial channel; Flutter mobile development follows once the API and core journeys are stable.
- The platform shall not complete vehicle purchase, escrow or ownership transfer in the first release.
- Public prices shall be described as advertised/asking prices, not official valuations.
- Private addresses, NIC numbers, password hashes, refresh tokens and internal moderation notes shall never be exposed publicly.
- Faceted search parameters shall not create an uncontrolled number of indexable URLs; curated landing pages are preferred.
- Core marketplace functionality shall be prioritized before valuation, auctions, AI features or complex financial integrations.

11. Current implementation allows up to 5 listing images at 5 MB each; the final commercial image policy is TBD.
12. Offset pagination may be used initially; cursor/keyset pagination is preferred when catalogue scale requires it.

2.6 Assumptions and Dependencies [\[Back to TOC\]](#)

ID	Statement	Type
AS-001	Sufficient sellers and dealers will provide quality inventory for marketplace launch.	Assumption
AS-002	Sri Lankan brand/model/category/district/city taxonomy can be maintained by administrators.	Assumption
AS-003	Transactional email and object-storage credentials will be configured in production.	Assumption
DE-001	Marketplace value depends on both listing supply and buyer traffic.	Dependency
DE-002	Organic acquisition depends on correct indexing of permanent SEO pages.	Dependency
DE-003	Dealer map discovery depends on valid location coordinates.	Dependency
DE-004	Future Flutter push notifications depend on a mobile push provider and user permission.	Dependency

3. System Features [\[Back to TOC\]](#)

3.1 Authentication and Role-Based Access [\[Back to TOC\]](#)

Priority: High

Description: Registers users, authenticates sessions and protects functions by role.

ID	Requirement	Status
AUT-001	Visitors shall register using first name, last name, email and password; phone may be optional.	Implemented
AUT-002	Users shall authenticate with email/password and receive access and refresh credentials.	Implemented
AUT-003	Refresh sessions shall support rotation and invalidation.	Implemented
AUT-004	Logout shall invalidate the relevant refresh session.	Implemented
AUT-005	Forgot-password and reset-password shall use time-limited tokens.	Implemented
AUT-006	Email verification and resend-verification shall be supported.	Implemented
AUT-007	The system shall distinguish buyer, seller, dealer, parts dealer and admin roles.	Implemented
AUT-008	RBAC shall prevent access to functions outside the user's permissions.	Implemented
AUT-009	Sensitive authentication endpoints shall be rate-limited.	Implemented
AUT-010	Phone verification via OTP shall be supported as a future account-trust control.	Partial

3.2 User Profile and Account [\[Back to TOC\]](#)

Priority: High

Description: Provides account details, avatar and user-specific marketplace tools.

ID	Requirement	Status
PRF-001	Authenticated users shall retrieve their profile.	Implemented
PRF-002	Authenticated users shall update permitted profile fields.	Implemented
PRF-003	Users shall upload/remove an avatar using object storage.	Implemented
PRF-004	Account navigation shall expose listings, favourites, saved searches, messages, notifications and role-specific tools.	Implemented
PRF-005	Users shall manage notification-channel preferences for in-app, email and future push.	Planned

3.3 Marketplace Taxonomy [\[Back to TOC\]](#)

Priority: High

Description: Maintains structured motorcycle, location and parts classifications.

ID	Requirement	Status
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ID	Requirement	Status
TAX-001	Public brand, model, category, district and city lists shall be available.	Implemented
TAX-002	Administrators shall manage brands, models, districts and cities.	Implemented
TAX-003	Administrators shall manage brand logos and category cover images.	Implemented
TAX-004	Bike models shall belong to brands and may belong to categories.	Implemented
TAX-005	Bike variants/grades shall be supported as a first-class taxonomy entity.	Planned
TAX-006	Part categories shall support parent/child hierarchy.	Implemented

3.4 Motorcycle Listing Management [\[Back to TOC\]](#)

Priority: High

Description: Captures structured bike data and manages a moderated listing lifecycle.

ID	Requirement	Status
LST-001	A seller shall create a listing with brand, model, category, district, city, title, description, LKR price, negotiable flag, manufacture year, mileage, fuel type, transmission, condition and contact phone.	Implemented
LST-002	A listing may include registration year, engine capacity, colour, WhatsApp and dealer association.	Implemented
LST-003	Dealer inventory may store internal cost price and purchase date for performance reporting.	Implemented
LST-004	Listing input shall be validated before persistence.	Implemented
LST-005	Sellers shall upload validated listing images to object storage.	Implemented
LST-006	New listings shall begin as draft and require submission for moderation before public activation.	Implemented
LST-007	Administrators shall approve or reject pending listings; rejection shall include a reason.	Implemented
LST-008	Owners shall edit, pause, resume and mark their own listings sold subject to status rules.	Implemented
LST-009	Automatic expiry and renew shall remove stale inventory and allow controlled republishing.	Planned
LST-010	Variant, service history, modifications, accident declaration, previous owners and finance/exchange fields shall be supported where approved for launch.	Planned

3.5 Search and Discovery [\[Back to TOC\]](#)

Priority: High

Description: Allows buyers to find relevant motorcycles quickly.

ID	Requirement	Status
SRH-001	Users shall browse active motorcycles and motorcycle category pages.	Implemented
SRH-002	Users shall filter by brand, model, category, district, price range, manufacture year and condition.	Implemented
SRH-003	Search shall later include variant, registration year, mileage, engine capacity, fuel, transmission, city, seller type, negotiable and featured filters.	Partial / Planned
SRH-004	Users shall sort by newest, oldest, price, mileage and year.	Implemented
SRH-005	Popularity / most-viewed sorting shall be supported.	Planned
SRH-006	Keyword search shall support common motorcycle queries; advanced synonyms/search-engine support may be added when required.	Partial
SRH-007	The home page shall expose search, categories, featured inventory and marketplace discovery.	Implemented
SRH-008	Permanent brand, model and location pages shall support SEO discovery.	Partial
SRH-009	Public listing URLs shall remain unique and stable.	Partial

3.6 Listing Detail, Contact and Engagement [\[Back to TOC\]](#)

Priority: High

Description: Presents a complete advertisement and connects the buyer to the seller.

ID	Requirement	Status
DET-001	Listing detail shall show price, negotiable status, specifications, location, seller type, gallery and description.	Implemented
DET-002	Guests shall be able to submit a contact inquiry.	Implemented
DET-003	Phone/WhatsApp visibility may be controlled by authentication or seller visibility rules.	Implemented
DET-004	Contact clicks shall be recorded for seller/dealer performance.	Implemented
DET-005	Listing views shall be recorded with abuse protection.	Implemented
DET-006	Related spare/modified parts may be surfaced through fitment.	Implemented
DET-007	Users shall share a listing using its public URL.	Implemented
DET-008	Registered users shall have a recently viewed history.	Planned

3.7 Favourites, Comparison and Saved Searches [\[Back to TOC\]](#)

Priority: Medium

Description: Supports buyer retention and repeat discovery.

ID	Requirement	Status
FAV-001	Registered users shall add/remove motorcycle favourites.	Implemented
FAV-002	Price reductions on favourited listings shall generate notifications where configured.	Implemented
CMP-001	Users shall compare up to three motorcycles side by side.	Implemented
SAV-001	Registered users shall save and reopen named search queries.	Implemented
SAV-002	Saved searches shall optionally alert users when matching new listings are published.	Partial
SAV-003	Part listings shall support favourites.	Implemented

3.8 Messaging and Notifications [\[Back to TOC\]](#)

Priority: Medium

Description: Supports direct marketplace communication and event alerts.

ID	Requirement	Status
MSG-001	Authenticated users shall start a conversation on a bike or part listing.	Implemented
MSG-002	Conversation participants shall reply within the thread.	Implemented
MSG-003	Conversation creation/reply endpoints shall be rate-limited.	Implemented
NTF-001	The system shall persist in-app notifications and unread state.	Implemented
NTF-002	Sellers shall receive approval/rejection and promotion decision notifications.	Implemented
NTF-003	Transactional email shall be sent through the configured provider.	Implemented
NTF-004	The Flutter application shall support push notifications when the mobile release is introduced.	Planned

3.9 Bike Dealer Ecosystem [\[Back to TOC\]](#)

Priority: High

Description: Provides approved motorcycle businesses with a branded marketplace presence.

ID	Requirement	Status
DLR-001	Registered users shall apply for a bike-dealer profile.	Implemented
DLR-002	Administrators shall approve, reject or suspend dealer applications.	Implemented
DLR-003	Approved dealers shall have public showroom pages identified by stable slugs.	Implemented
DLR-004	Dealers shall manage inventory and relevant vehicle documents with expiry information where supported.	Implemented
DLR-005	Dealers shall upload showroom images and optional map coordinates.	Implemented
DLR-006	Public dealer directory and map pages shall list approved dealers.	Implemented
DLR-007	Dealer listings shall be visibly distinguished from private-seller listings.	Implemented
DLR-008	Dealers shall view listing/lead performance.	Implemented
DLR-009	Multi-user dealer organizations shall be supported in a later release.	Future
DLR-010	Paid dealer subscription packages and listing caps shall be supported in a later commercial release.	Future

3.10 Spare and Modified Parts Marketplace [\[Back to TOC\]](#)

Priority: Medium-High

Description: Creates a motorcycle-parts ecosystem connected to bike fitment.

ID	Requirement	Status
PRT-001	Part listings shall distinguish spare and modified kinds.	Implemented
PRT-002	Approved parts dealers shall create, edit, submit, pause and mark part listings.	Implemented
PRT-003	Part listings shall support hierarchical categories and brand/model fitment.	Implemented
PRT-004	Public pages shall exist for all parts, spare parts, modified parts and category pages.	Implemented
PRT-005	Users shall inquire on parts and engagement shall be measurable.	Implemented
PRT-006	Administrators shall moderate part listings and categories.	Implemented
PRT-007	Parts dealers shall view parts performance.	Implemented

3.11 Parts Dealer Ecosystem [\[Back to TOC\]](#)

Priority: Medium

Description: Supports approved parts businesses.

ID	Requirement	Status
PDL-001	A user shall apply for a parts-dealer profile.	Implemented
PDL-002	Administrators shall approve or reject parts-dealer applications.	Implemented
PDL-003	Approved parts dealers shall have public showrooms and directory/map presence.	Implemented

3.12 Reports, Trust and Moderation [\[Back to TOC\]](#)

Priority: High

Description: Protects marketplace quality and enables operational intervention.

ID	Requirement	Status
RPT-001	Registered users shall report suspicious bike or part listings.	Implemented
RPT-002	Report creation shall be rate-limited.	Implemented
RPT-003	Administrators shall resolve reports with an action/note and may take content down.	Implemented
RPT-004	Sensitive private/internal information shall never appear on public pages.	Implemented
RPT-005	Automatic expiry shall remove stale inventory from active search.	Planned
RPT-006	Duplicate-image and fraud-risk signals may assist moderators in future releases.	Future
RPT-007	CAPTCHA may protect high-abuse endpoints when traffic justifies it.	Future

3.13 Promotions and Homepage Advertising [\[Back to TOC\]](#)

Priority: Medium

Description: Supports initial marketplace monetization.

ID	Requirement	Status
PRO-001	The system shall define promotion packages for eligible bike and part listings.	Implemented
PRO-002	Sellers shall request homepage placement by selecting a package and uploading a bank-payment slip.	Implemented
PRO-003	Administrators shall approve/reject promotion requests and manage placements/bank details.	Implemented
PRO-004	Approved placements shall render in configured homepage promotional positions.	Implemented
PRO-005	Online paid checkout for featured/top/urgent products shall be supported in a later release.	Future
PRO-006	Dealer subscriptions and brand advertising with online payment shall be supported later.	Future

3.14 Administration Console [\[Back to TOC\]](#)

Priority: High

Description: Provides operational control over marketplace content and users.

ID	Requirement	Status
ADM-001	Administrators shall view a marketplace overview dashboard.	Implemented
ADM-002	Administrators shall manage users, bike listings, part listings, dealers and parts dealers.	Implemented
ADM-003	Administrators shall manage marketplace taxonomy and visual assets.	Implemented
ADM-004	Administrators shall review reports and homepage promotions.	Implemented
ADM-005	Buying guides/blog content may be moved to an administrator-managed CMS.	Planned
ADM-006	Sensitive administrator actions shall be retained in an audit log.	Planned

3.15 SEO and Public Content [\[Back to TOC\]](#)

Priority: High

Description: Makes the marketplace discoverable through search engines.

ID	Requirement	Status
SEO-001	Important public pages shall be server-rendered or statically generated rather than client-only.	Implemented
SEO-002	Important pages shall emit title, meta description, canonical URL and Open Graph metadata.	Implemented

ID	Requirement	Status
SEO-003	Listing pages shall emit structured data consistent with visible content and shall not fabricate ratings.	Implemented
SEO-004	XML sitemaps shall include approved indexable public pages for supported locales.	Partial
SEO-005	Production shall be indexable while staging/non-production environments shall be protected from indexing.	Partial
SEO-006	Faceted search URLs shall use canonical/noindex controls to prevent index explosion.	Partial
SEO-007	Buying guides shall be available as public content pages.	Implemented
SEO-008	Public locale routing shall support /en and /si.	Implemented

3.16 Platform Services [\[Back to TOC\]](#)

Priority: High

Description: Provides health, storage, errors and background-processing capabilities.

ID	Requirement	Status
PLT-001	The API shall expose a health endpoint covering service/database status and monitoring integration.	Implemented
PLT-002	Image uploads shall validate file type/size and use object storage.	Implemented
PLT-003	API errors shall use a consistent success/error response envelope.	Implemented
PLT-004	The API shall use security headers, controlled CORS and hide framework disclosure headers.	Implemented
PLT-005	Structured error monitoring shall be available for web and API.	Implemented
PLT-006	Saved-search matching, listing expiry and queued notifications shall run through background workers when introduced.	Planned

4. External Interface Requirements [\[Back to TOC\]](#)

4.1 User Interfaces [\[Back to TOC\]](#)

ID	Requirement	Status
UI-001	The public interface shall be mobile-first and usable on phone, tablet and desktop.	Implemented
UI-002	Primary navigation shall provide search/browse, sell, authentication and locale switching.	Implemented
UI-003	Listings, dealers, parts and account screens shall use consistent cards, forms, validation and empty states.	Implemented
UI-004	The administrator interface shall be a separate operational console.	Implemented
UI-005	Price copy shall consistently use asking/advertised price terminology.	Implemented
UI-006	English and Sinhala experiences shall provide equivalent core journey coverage.	Partial
UI-007	Key journeys shall target WCAG 2.1 AA accessibility for labels, keyboard use, focus, contrast and alternative text.	Partial
UI-008	The future Flutter application shall follow the same product roles, business rules and content terminology as the web platform.	Planned

4.2 Hardware Interfaces [\[Back to TOC\]](#)

No specialized user hardware is required. Buyers and sellers use standard smartphones, tablets or computers. Production infrastructure may run on a VPS or equivalent cloud compute; final production sizing is a deployment decision.

4.3 Software Interfaces [\[Back to TOC\]](#)

ID	Interface	Requirement	Status
SR-001	Next.js ↔ NestJS	Web shall consume the versioned /api/v1 REST API.	Implemented
SR-002	NestJS ↔ PostgreSQL	Operational data shall be persisted through the ORM and migrations.	Implemented
SR-003	NestJS ↔ Redis	Cache/coordination shall be used when Redis is configured.	Implemented
SR-004	NestJS ↔ Object Storage	Marketplace media shall use an S3-compatible API.	Implemented
SR-005	NestJS ↔ Email Provider	Transactional email shall use the configured provider.	Implemented
SR-006	Web/API ↔ Monitoring	Application errors shall be reportable to the configured monitoring platform.	Implemented / Partial

ID	Interface	Requirement	Status
SR-007	Flutter ↔ NestJS	The planned Flutter app shall reuse /api/v1 and the same business rules.	Planned
SR-008	Future Search Engine	A dedicated search service may replace/augment PostgreSQL search if scale requires it.	Future

4.4 Communications Interfaces [\[Back to TOC\]](#)

ID	Requirement	Status
CI-001	Production client-to-server traffic shall use HTTPS.	Planned for production
CI-002	Application data shall use documented REST/JSON requests and consistent response envelopes.	Implemented
CI-003	Authentication credentials and tokens shall be transmitted only through protected API channels.	Implemented
CI-004	CORS shall allow only configured trusted origins.	Implemented
CI-005	The Flutter client shall communicate through the same HTTPS REST API; direct database access is prohibited.	Planned

5. Nonfunctional Requirements [\[Back to TOC\]](#)

5.1 Performance and Availability [\[Back to TOC\]](#)

ID	Requirement	Status
PE-001	Important marketplace pages shall render meaningful first content server-side or statically where appropriate.	Implemented
PE-002	Upload size/type controls shall protect media payload and storage.	Implemented
PE-003	Public listing queries shall use database indexes appropriate to status, date, brand, location, category and price.	Implemented
PE-004	Public list endpoints shall enforce bounded page sizes.	Implemented
PE-005	Home, search and listing pages shall have measured mobile performance budgets; final numeric thresholds are TBD.	TBD
PE-006	Early production availability target is 99.5% or better, subject to final infrastructure design.	Planned
PE-007	Large catalogues shall migrate from expensive offsets to cursor/keyset pagination when required.	Planned

5.2 Security and Privacy [\[Back to TOC\]](#)

ID	Requirement	Status
SE-001	Production traffic shall be HTTPS-only.	Planned for production
SE-002	Passwords shall be stored using bcrypt or an equivalent approved password-hashing algorithm.	Implemented
SE-003	Authentication shall use short-lived access tokens and rotatable refresh sessions.	Implemented
SE-004	Mutating APIs shall validate input using shared validation schemas.	Implemented
SE-005	Role-based guards shall protect administrator and role-restricted functions.	Implemented
SE-006	Rate limits shall protect authentication, upload, contact, report, messaging and write routes.	Implemented
SE-007	Security headers and controlled CORS shall be enabled; unnecessary framework headers shall be disabled.	Implemented
SE-008	Secrets shall be stored outside source control in environment/secret management.	Implemented as process
SE-009	Image uploads shall reject unsupported types and oversized files.	Implemented
SE-010	Database queries shall remain parameterized through the ORM.	Implemented
SE-011	Production and staging configuration shall be separated.	Partial
SE-012	Public endpoints shall expose only the minimum personal data required for marketplace contact.	Implemented / policy
SE-013	Administrator audit logging shall record sensitive operational actions.	Planned

5.3 Software Quality Attributes [\[Back to TOC\]](#)

Attribute	Requirement	Status
Usability	Core buyer/seller/dealer journeys shall be easy to complete on mobile without specialist knowledge.	Partial - formal usability testing pending
Maintainability	Shared TypeScript contracts, linting, modular services and migrations shall support maintainable development.	Implemented
Reliability	Illegal listing-state transitions shall be rejected; health checks shall expose service/database status.	Implemented

Attribute	Requirement	Status
Testability	Business rules, services, API flows and validation shall support unit, integration and end-to-end testing.	Partial
Accessibility	Core web journeys shall target WCAG 2.1 AA.	Partial
Scalability	API shall remain stateless where practical; PostgreSQL is the source of truth and Redis is optional acceleration.	Implemented as architecture
Observability	Health, structured logs and error monitoring shall support production diagnosis.	Partial
Internationalization	English and Sinhala routing/content shall be supported; full Sinhala coverage is a launch decision.	Partial
SEO quality	Public pages shall remain crawlable, canonical and semantically structured without index spam.	Partial

6. Business Rules [\[Back to TOC\]](#)

ID	Business Rule	Status
BSR-001	Bike and part listings shall not appear in public search until approved and active.	Implemented
BSR-002	Only the listing owner or an authorized administrator shall modify listing lifecycle state.	Implemented
BSR-003	Dealer and parts-dealer profiles shall not become public before administrator approval.	Implemented
BSR-004	Private sellers shall not attach listings to dealer profiles they do not control.	Implemented
BSR-005	A dealer shall be approved before listings can be attached to that dealer.	Implemented
BSR-006	Motorcycle purchase payment and ownership transfer occur outside ThrottleLK in the first release.	Implemented / scope
BSR-007	Sold listings shall be retained rather than automatically hard-deleted; public presentation may change.	Implemented
BSR-008	Administrator rejection/take-down shall record a reason and notify the affected seller where supported.	Implemented
BSR-009	Sensitive personal/security/internal moderation data shall not be exposed publicly.	Implemented
BSR-010	Manual homepage promotions shall go live only after administrator review of the payment evidence.	Implemented
BSR-011	Advertised prices are asking prices and shall not be represented as official valuations.	Implemented / policy
BSR-012	ThrottleLK shall not claim automated vehicle valuation unless a separate validated valuation product is introduced.	Implemented / scope
BSR-013	Future automated paid promotion and subscription products shall follow approved commercial policy before release.	Future

7. Data Requirements [\[Back to TOC\]](#)

7.1 Operational Data Groups [\[Back to TOC\]](#)

Domain	Core entities / concepts	Status
Identity & access	users, roles, user_roles, refresh sessions, verification/reset tokens	Implemented
Taxonomy	brands, bike models, categories, districts, cities, part categories; bike variants planned	Mixed
Bike marketplace	listings, listing images, inquiries, engagement events, favourites, saved searches	Implemented

Domain	Core entities / concepts	Status
Dealers	dealers, dealer images, inventory items, inventory documents	Implemented
Parts	part listings, images, fitments, favourites, inquiries, engagement, parts dealers	Implemented
Trust & communication	reports, notifications, conversations, messages	Implemented
Commercial growth	promotion packages, requests, bank accounts/settings, homepage placements	Implemented
Operations	administrator audit log	Planned

7.2 Motorcycle Listing Data [\[Back to TOC\]](#)

Field group	Examples	Requirement state
Identity/classification	brand, model, category, district, city	Required
Commercial	title, description, LKR price, negotiable	Required
Vehicle	manufacture year, mileage, fuel, transmission, condition	Required
Contact	phone; optional WhatsApp	Required / Optional
Additional	registration year, engine capacity, colour, dealer association	Optional
Dealer internal	cost price, purchase date	Optional / restricted
Planned detail	variant, service history, modifications, accident declaration, previous owners, finance/exchange	Planned

7.3 Data Integrity [\[Back to TOC\]](#)

ID	Requirement	Status
DPR-001	Public listing slugs shall be unique.	Implemented
DPR-002	Soft-deleted/removed listings shall be excluded from normal public queries.	Implemented
DPR-003	Search indexes shall support high-value query dimensions.	Implemented
DPR-004	Large catalogue pagination shall migrate to cursor/keyset when required.	Planned
DPR-005	Duplicate listing/image detection may be introduced to strengthen moderation.	Future
DPR-006	The Flutter app shall not maintain a separate authoritative business database.	Planned / architecture

8. Other Requirements and Scope Boundaries [\[Back to TOC\]](#)

8.1 Data Storage [\[Back to TOC\]](#)

PostgreSQL is the source of truth for operational marketplace data. Binary media is stored in object storage and referenced by key/URL. Redis is an acceleration/coordination layer, not a system of record. The future Flutter application shall reuse the same backend and authoritative data.

8.2 Backup, Retention and Recovery [\[Back to TOC\]](#)

Final retention periods, purge rules, backup frequency, Recovery Time Objective (RTO) and Recovery Point Objective (RPO) remain TBD. These values must be approved before production operations are considered fully baselined.

8.3 Legal, Privacy and Responsible Marketplace Requirements [\[Back to TOC\]](#)

ID	Requirement	Status
LEG-001	Personal data shall be limited to information needed for account, marketplace and contact journeys.	Implemented as design
LEG-002	Seller contact data shall be disclosed only according to product visibility rules.	Implemented
LEG-003	ThrottleLK shall not present itself as an official valuation authority or guarantee sale prices.	Policy
LEG-004	Promotion pricing and dealer packages shall be approved as commercial products before public sale.	Planned
LEG-005	Future finance/insurance lead products shall be introduced only after applicable business/legal review.	Future
LEG-006	Terms of service, privacy notice and marketplace rules shall be published before full public commercial launch.	Planned

8.4 Release Scope Priorities [\[Back to TOC\]](#)

Priority	Functions	Current position
Core / MVP	Authentication, taxonomy, bike listings, images, search/filter, listing detail, seller management, moderation, responsive web and core SEO.	Largely implemented; remaining gaps include richer filters, expiry/renew, final image policy and some SEO rules.
Secondary	Favourites, compare, saved searches, messaging, notifications, dealer showrooms, parts marketplace, maps, analytics and promotions.	Mostly implemented; alert workers, preferences, recently viewed, full Sinhala and some analytics remain.
Mobile expansion	Flutter buyer/seller/dealer application using the same API.	Planned after core web/API stability.
Future commercial	Online promotion checkout, dealer subscriptions, advertising products, finance/insurance leads.	Future.
Future advanced	Valuation, inspection, escrow, auctions, AI assistance, fraud intelligence, dedicated search engine and dealer ERP/team accounts.	Outside first release.

9. Detailed Use Case Specifications [\[Back to TOC\]](#)

UC-01 Register and Authenticate [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Guest / Registered User
Preconditions	Web/API reachable
Main flow	User registers or logs in; system validates credentials, creates/uses session tokens and grants access according to role.
Alternative / exception	Invalid data, invalid credentials, expired token or rate limit produces a controlled error.
Postconditions	Authenticated session exists.
Status	Implemented

UC-02 Browse and Search Motorcycles [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Guest / Registered Buyer
Preconditions	Active listings exist
Main flow	User opens marketplace/SEO page, applies filters/sort and opens a listing from active results.
Alternative / exception	No results shows an empty state; planned filters are unavailable until implemented.
Postconditions	Relevant result set or empty state displayed.
Status	Partial

UC-03 View Listing and Contact Seller [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Guest / Registered Buyer
Preconditions	Public active listing exists
Main flow	User reviews gallery/specs/seller, then calls, opens WhatsApp, submits inquiry or starts authenticated messaging; engagement is recorded where applicable.
Alternative / exception	Visibility rules may require login; own listing cannot be contacted as another buyer.
Postconditions	Seller receives a lead or conversation.
Status	Implemented

UC-04 Create and Submit Bike Listing [\[Back to TOC\]](#)

Item	Specification
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Item	Specification
Primary actor(s)	Private Seller / Approved Dealer
Preconditions	Authenticated user; taxonomy available
Main flow	Seller completes listing, validates data, creates draft, uploads images and submits for review; admin approval activates it.
Alternative / exception	Validation failure returns field errors; rejection stores a reason.
Postconditions	Listing is draft, pending, active or rejected.
Status	Implemented

UC-05 Moderate Listing / Dealer [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Administrator
Preconditions	Admin authenticated
Main flow	Admin reviews pending content/application and approves, rejects, suspends or takes down according to policy.
Alternative / exception	Invalid transition is rejected; reason may be required.
Postconditions	Marketplace state and notification updated.
Status	Implemented

UC-06 Favourite, Compare and Save Search [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Registered Buyer; compare may be guest-local
Preconditions	Listings exist
Main flow	User favourites a bike, compares up to three bikes and saves a named search.
Alternative / exception	Guests cannot persist favourites/searches; saved-search alerts are not fully automated yet.
Postconditions	Buyer retention data persists.
Status	Partial

UC-07 Apply as Bike Dealer [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Seller / Dealer Applicant + Administrator
Preconditions	Registered user
Main flow	User submits business profile; admin reviews; approved dealer configures showroom/map/inventory and publishes dealer-linked listings.
Alternative / exception	Rejection stores a reason; unapproved dealer cannot publish as dealer.
Postconditions	Approved public dealer showroom exists.
Status	Implemented

UC-08 List Spare / Modified Part [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Approved Parts Dealer + Administrator
Preconditions	Approved parts dealer
Main flow	Dealer creates part with category/fitment/images, submits, admin approves and listing becomes searchable.
Alternative / exception	Invalid fitment/validation is rejected.
Postconditions	Active part listing exists.
Status	Implemented

UC-09 In-Platform Messaging [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Registered Buyer + Listing Owner
Preconditions	Both authenticated; listing exists

Item	Specification
Main flow	Buyer starts conversation; owner replies; both access thread from account area.
Alternative / exception	Rate limits or own-listing rules may reject action.
Postconditions	Conversation persists.
Status	Implemented

UC-10 Request Homepage Promotion [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Seller / Dealer + Administrator
Preconditions	Eligible active listing; promotion package configured
Main flow	Seller selects package, uploads payment evidence, admin reviews and approved placement is scheduled.
Alternative / exception	Rejection records reason; no card checkout in current flow.
Postconditions	Approved listing may appear in promotional slot.
Status	Implemented - manual settlement

UC-11 Report a Listing [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Registered User + Administrator
Preconditions	Authenticated user; target listing exists
Main flow	User submits report; admin reviews, resolves and may take content down.
Alternative / exception	Rate limit or invalid report rejected.
Postconditions	Report resolved; content may change status.
Status	Implemented

UC-12 Use Flutter Mobile Application [\[Back to TOC\]](#)

Item	Specification
Primary actor(s)	Mobile User
Preconditions	Stable /api/v1; Flutter app released
Main flow	App authenticates and uses the same API for browse, favourites, messaging, seller and dealer journeys.
Alternative / exception	Unavailable until mobile release; push permission may be denied.
Postconditions	Mobile and web share one backend/business truth.
Status	Planned

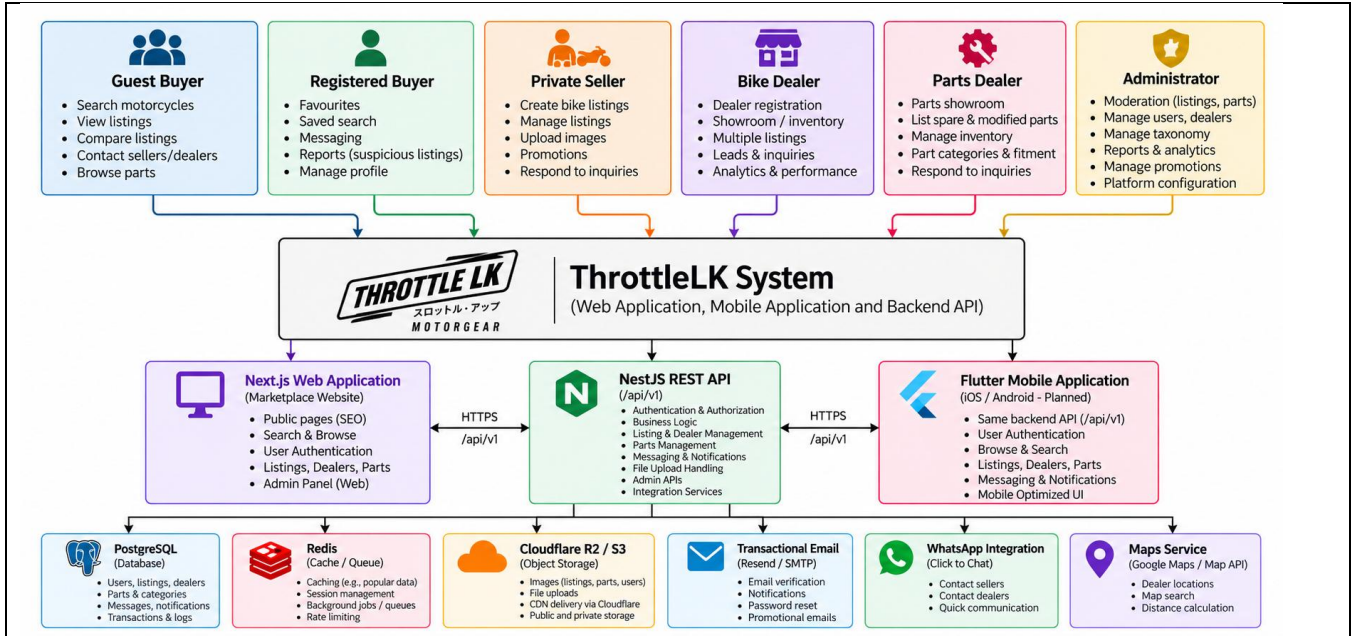
10. Requirements Traceability Matrix [\[Back to TOC\]](#)

Requirement group	Use case(s)	Primary module	Delivery	Verification family
AUT / PRF	UC-01	Authentication / Users	Implemented / Partial	TC-AUT / TC-PRF
TAX / SRH	UC-02	Taxonomy / Public Search	Mixed	TC-TAX / TC-SRH
LST	UC-04, UC-05	Listings / Images / Moderation	Mixed	TC-LST
DET	UC-03	Listing Detail / Inquiry / Engagement	Mixed	TC-DET
FAV / CMP / SAV	UC-06	Buyer Retention	Mixed	TC-FAV / TC-CMP / TC-SAV
MSG / NTF	UC-09	Conversations / Notifications	Mixed	TC-MSG / TC-NTF
DLR	UC-07	Dealer / Inventory	Mixed	TC-DLR
PRT / PDL	UC-08	Parts / Parts Dealers	Implemented	TC-PRT / TC-PDL
RPT	UC-11	Reports / Moderation	Mixed	TC-RPT
PRO	UC-10	Promotions	Mixed	TC-PRO
ADM	UC-05	Admin Console	Mixed	TC-ADM
SEO	UC-02, UC-03	Web Metadata / Sitemap / Robots	Partial	TC-SEO
SE / PE / UI	All relevant use cases	Cross-cutting quality	Mixed	TC-SE / TC-PE / TC-UI

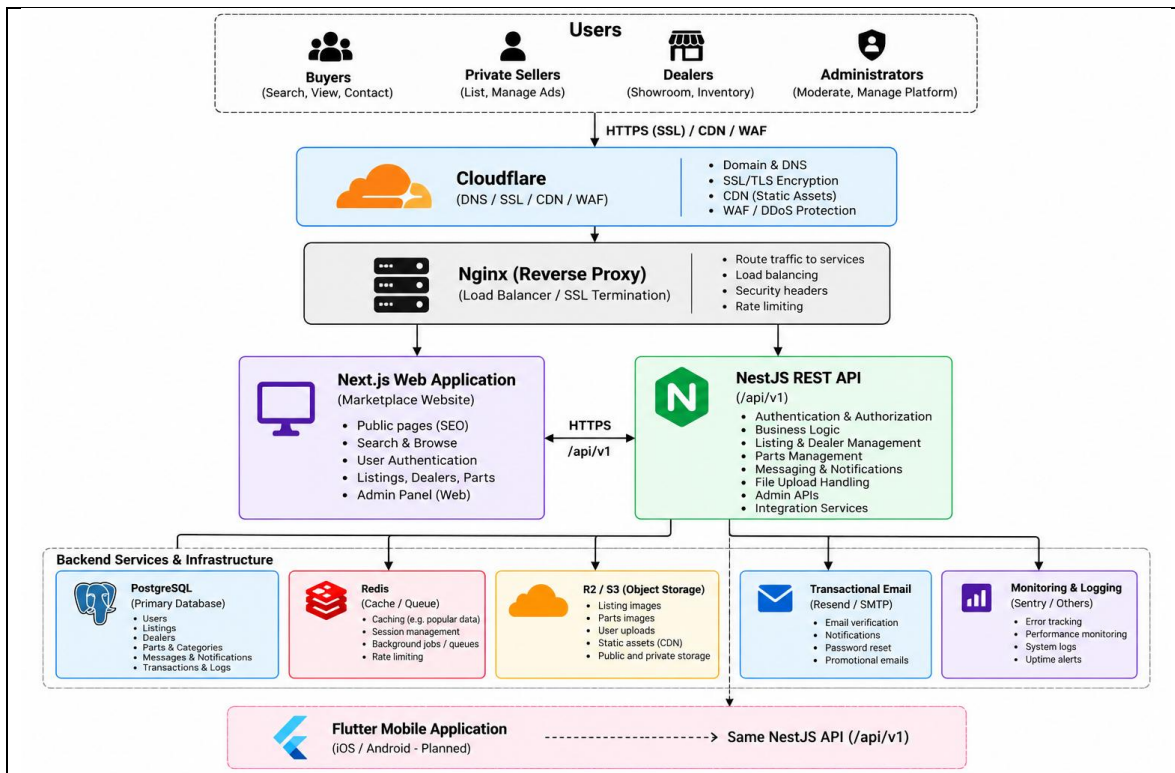
Requirement group	Use case(s)	Primary module	Delivery	Verification family
OE-010 / SR-007 / CI-005	UC-12	Flutter Mobile Client	Planned	TC-MOB

11. Analysis and Architecture Models [\[Back to TOC\]](#)

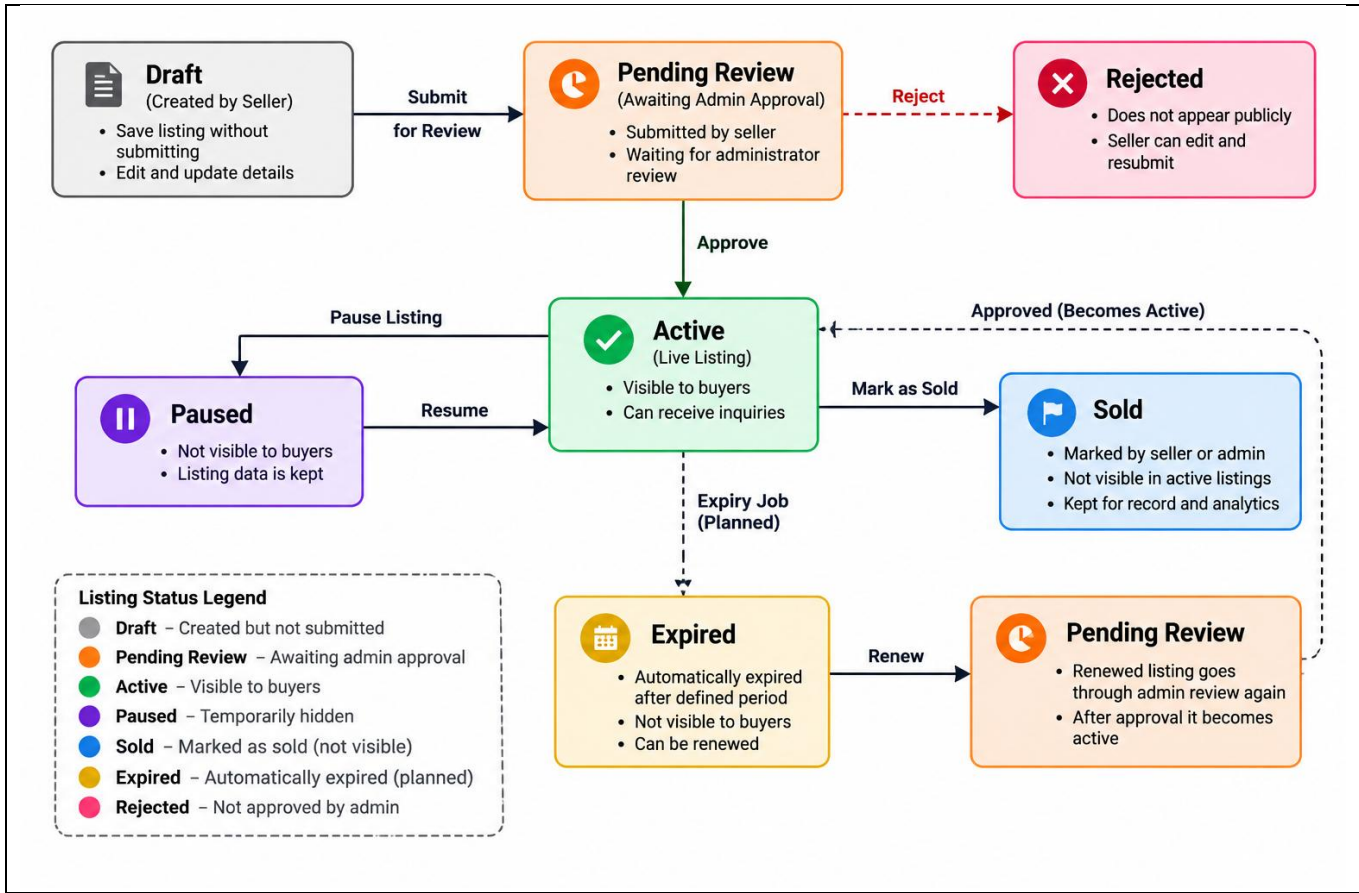
11.1 System Context / Use Case View [\[Back to TOC\]](#)



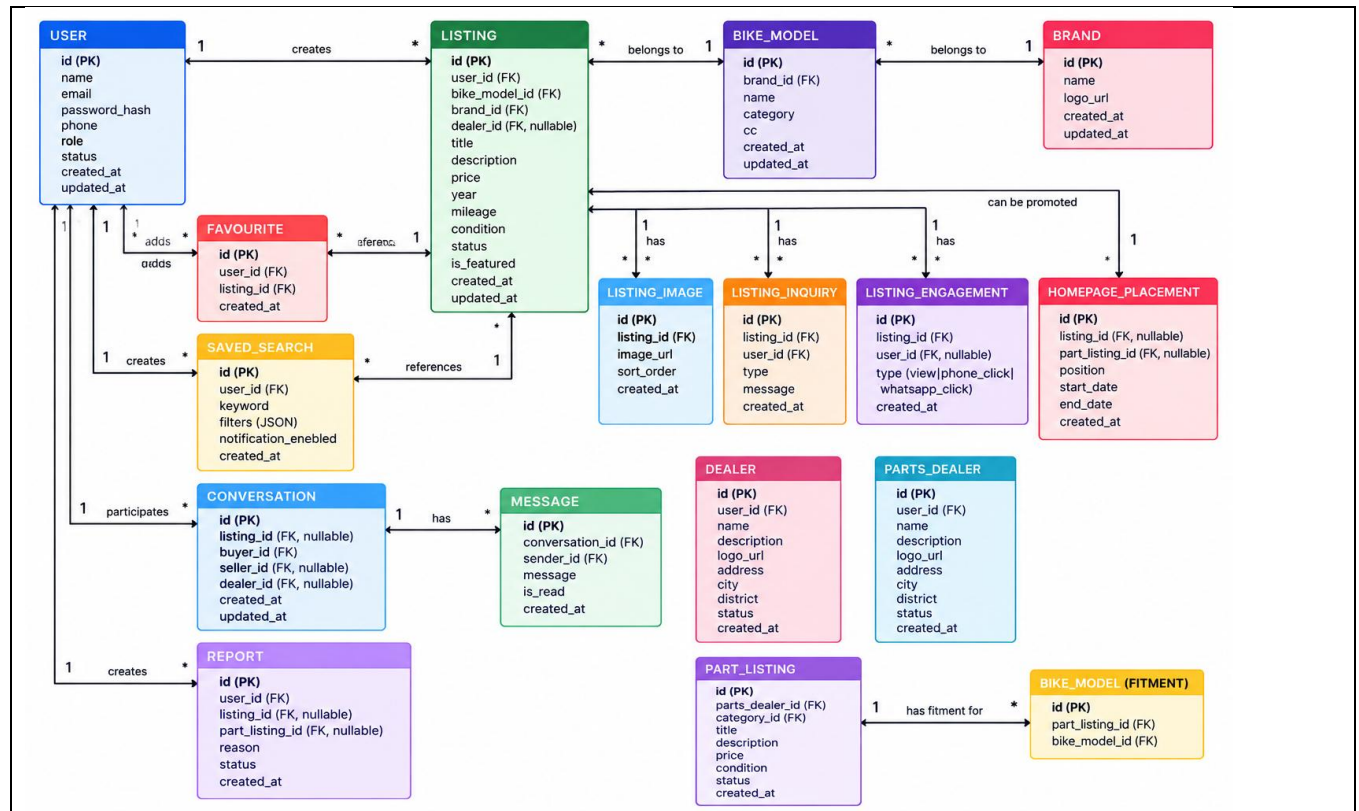
11.2 System Architecture [\[Back to TOC\]](#)



11.3 Listing Lifecycle [Back to TOC]



11.4 Conceptual Data Relationships [Back to TOC]



12. Implementation Gap and Release Readiness [\[Back to TOC\]](#)

12.1 Already Available [\[Back to TOC\]](#)

- Next.js web + NestJS API + PostgreSQL marketplace foundation.
- Authentication, roles, email verification and password recovery.
- Bike listing lifecycle with images and administrator moderation.
- Core browse/search, listing detail, inquiry, phone/WhatsApp and engagement tracking.
- Bike dealer and parts dealer applications, showrooms, maps and inventory.
- Spare/modified parts marketplace with fitment.
- Favourites, compare, saved-search storage, conversations and notification feed.
- Reports, homepage promotions and administration console.
- SEO foundation including SSR, metadata, canonicals, structured data, sitemap and robots.
- English/Sinhala locale routing and production-oriented security middleware.

12.2 Partial / Launch-Hardening Work [\[Back to TOC\]](#)

Area	Current state	Remaining work
Search	Core filters and sorts exist.	Add agreed advanced filters, seller type, variant and popularity sort.
Saved-search alerts	Searches can be stored.	Implement matching/background notification worker.
Sinhala	Locale route and partial dictionary exist.	Complete core commercial journeys and QA.
SEO	Core metadata/URLs exist.	Finalize faceted noindex/canonical policy and remaining landing-page patterns.
Images	Current max 5 images.	Approve final commercial image count/size policy.
Listing expiry	Status concept exists.	Implement scheduler, active-search removal and renew flow.
Notifications	In-app/email capability exists.	Add user preference centre; Flutter push later.
Accessibility	Some semantic controls exist.	Perform WCAG 2.1 AA audit on key journeys.
Performance	Architecture supports optimization.	Define measurable mobile SLOs and load-test baseline.
Operations	App is deployable.	Finalize hosting, backups, RTO/RPO, monitoring and runbooks.
Auditability	Moderation exists.	Add administrator audit log for sensitive actions.

12.3 Planned / Future [\[Back to TOC\]](#)

- Flutter mobile application using the same NestJS REST API.
- Online promotion checkout, dealer subscriptions and commercial advertising products.
- Phone OTP verification and recently viewed history.
- Dealer team accounts and ERP-style integrations.
- Finance/insurance lead integrations subject to business/legal review.
- AI listing assistance, fraud/duplicate-image intelligence and advanced recommendations.
- Dedicated search engine if PostgreSQL search becomes insufficient.
- Vehicle valuation, inspection, escrow and auctions only as separate future product decisions.

13. Finalization Notes [\[Back to TOC\]](#)

This SRS intentionally keeps unresolved commercial and operational decisions as TBD items rather than inventing values. Before the production launch baseline is approved, the Product Owner and engineering team should close the open issues in Appendix B and convert launch-critical TBDs into measurable acceptance criteria.

Requirements marked Implemented should be regression-tested before each release. Partial and Planned requirements are mandatory only when they belong to the agreed release scope. Future requirements remain roadmap items and should not block the first commercial launch unless the Product Owner explicitly promotes them into scope.

Appendix A: Glossary [\[Back to TOC\]](#)

Term	Definition
API	Application Programming Interface; ThrottleLK's backend REST surface is versioned under /api/v1.
Asking Price	The advertised price entered by a seller; not a confirmed transaction price or official valuation.
CDN	Content Delivery Network.
Fitment	Relationship indicating which motorcycle brand/model a part is intended to suit.
Flutter	Google's cross-platform UI framework using Dart; selected for the planned ThrottleLK mobile app.
JWT	JSON Web Token used as a short-lived access credential.
LKR	Sri Lankan Rupee.
Listing	A motorcycle or part advertisement with a controlled lifecycle status.
MVP	Minimum Viable Product; the core discovery-to-contact marketplace.
RBAC	Role-Based Access Control.
SEO	Search Engine Optimization.
Showroom	Public dealer or parts-dealer profile page.
Slug	URL-safe identifier used for public marketplace pages.
WAF	Web Application Firewall.
YOM	Year of Manufacture.

Appendix B: Open Issues / TBD Register [\[Back to TOC\]](#)

ID	Open decision
TBD-001	Exact set of launch-blocking advanced search filters.
TBD-002	Listing auto-expiry period and renew workflow.
TBD-003	Final image policy: count, per-file size and recommended photo set.
TBD-004	Final production hosting provider and Cloudflare/Nginx topology.
TBD-005	Numeric performance targets for LCP, API latency, concurrency and throughput.
TBD-006	Retention, purge, backup frequency, RTO and RPO.
TBD-007	Final public listing URL strategy and migration policy if changed.
TBD-008	Faceted-search canonical/noindex rules.
TBD-009	Minimum Sinhala completeness required for commercial launch.
TBD-010	Promotion prices, dealer subscription tiers and advertising packages.
TBD-011	Production transactional-email domain/from-address.
TBD-012	Whether saved-search alerts are launch-required or immediate post-launch.
TBD-013	Flutter mobile release scope: buyer-only first or buyer + seller/dealer journeys.
TBD-014	Terms of service, privacy notice, seller rules and moderation policy approval.
TBD-015	Production analytics/KPI definitions for leads, active listings, dealer performance and conversion.

END OF SOFTWARE REQUIREMENTS SPECIFICATION